

Evaluating Frozen and Antibody-Adapted ESM3 Representations for HIV Broadly Neutralizing Antibody Modeling

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Abstract

We evaluate whether representations from the ESM3 protein foundation model transfer to sequence-only modeling of HIV broadly neutralizing antibodies (bnAbs), and if domain adaptation via parameter-efficient fine-tuning alters this transfer. We partition heavy-chain sequences into lineage groups and evaluate frozen ESM3 and domain-adapted representations on classification and regression tasks. Our results demonstrate that frozen ESM3 representations effectively predict bnAb neutralization breadth and classify broad versus narrow neutralizers, outperforming handcrafted baseline features. However, unsupervised domain adaptation of the foundation model on antibody repertoires does not yield significant performance improvements. We also evaluate representation interpolation along lineage maturation pathways, showing that raw embedding-space interpolation is comparable to specialized variational autoencoder paths. These findings support frozen ESM3 as a viable exploratory representation for antibody modeling, while highlighting the importance of robust validation controls to avoid overstating performance gains.

1. Introduction

HIV-1 broadly neutralizing antibodies (bnAbs) are specialized immune system proteins capable of neutralizing a wide range of viral variants by binding to comparatively conserved regions on the viral envelope glycoprotein [19, 28]. Unlike ordinary antibodies, bnAbs develop over years through a process called somatic hypermutation, where B-cell receptor genes undergo iterative mutations and selection for increasing binding affinity.[add cite] This process yields complex clonal lineages (Figure 1 shows an example maturation pathway)—groups of antibodies that share a common genetic ancestor (the Unmutated Common Ancestor, or UCA) but diverge in sequence and function as they mature. Predicting neutralization breadth (the pro-

A									
Name	Putative V gene	Putative J gene	CDR3 length (aa)	CDR3 Sequence (aa)	V _H J _H % Mut (nt)	V _H J _H % Mut (aa)	Insertions/Deletions (aa)		
3H	IGHV4-59	IGHJ6*03	26	CARAQQGKRIVGVSFGFFYYNDVW	5%	11%			
4H				CARAQQGKRIVGVSFGELFYNDVW	6%	12%			
6H				CARAQRAKRIYGTISLGNFFDLYNDVW	19%	28%			
9H				CARTQHGRIYGVAFNEMFTFYNDVW	12%	18%			
10H				CATTGHRRIYGVAFNEMFTFYNDVW	16%	25%			
13H				CATTGHRRIYGVAFNEMFTFYNDVW	22%	27%			
22H				CATTGHRRIYGVAFNEMFTFYNDVW	23%	31%			
32H				CATTGHRRIYGVAFNEMFTFYNDVW	11%	17%			
36H				CATTGHRRIYGVAFNEMFTFYNDVW	15%	22%			
PGT121				CARTLHGRIYGVAFNEMFTFYNDVW	18%	23%			
PGT122				CATTGHRRIYGVAFNEMFTFYNDVW	19%	25%			
PGT123				CARALHGRIYGVAFNEMFTFYNDVW	20%	27%			
PGT124				CATTGHRRIYGVAFNEMFTFYNDVW	15%	21%			
PGT133				CARAQQGKRIVGVSFGFFYYNDVW	17%	25%			
PGT134				CARAQQGKRIVGVSFGELFYNDVW	17%	24%			

B									
Name	Putative V gene	Putative J gene	CDR3 length (aa)	CDR3 Sequence (aa)	V _H J _H % Total Mut (nt)	V _H J _H % Total Mut (aa)	Insertions/Deletions (aa)		
3L	IGLV3-21	IGLJ3*02	15	DYTCMWDSSRLNWF	9%	17%	+3 (FR3)		
4L				DYTCMWDSSRPTKWF	16%	29%	-7 (FR1)/ +3 (FR3)		
7L				DYTCMWDSSRPTKWF	21%	35%	-7 (FR1)/ +3 (FR3)		
8L				ATYCHMWDSSRPTKWF	22%	36%	-7 (FR1)/ +3 (FR3)		
64L				ATYCHMWDSSRPTKWF	22%	36%	-7 (FR1)/ +3 (FR3)		
80L				ATYCHMWDSSRPTKWF	19%	28%	-9 (FR1)/ +3 (FR3)		
84L				DYTCMWDSSRPTKWF	16%	28%	-4 (FR1)/ +3 (FR3)		
86L				DYTCMWDSSRPTKWF	21%	35%	-4 (FR1)/ +3 (FR3)		
87L				DYTCMWDSSRPTKWF	21%	33%	-4 (FR1)/ +3 (FR3)		
109L				DYTCMWDSSRPTKWF	17%	34%	-4 (FR1)/ +3 (FR3)		
PGT121				DYTCMWDSSRPTKWF	22%	28%	-7 (FR1)/ +3 (FR3)		
PGT122				DYTCMWDSSRPTKWF	18%	26%	-7 (FR1)/ +3 (FR3)		
PGT123				DYTCMWDSSRPTKWF	22%	34%	-7 (FR1)/ +3 (FR3)		
PGT124				DYTCMWDSSRPTKWF	19%	35%	-4 (FR1)/ +3 (FR3)		
PGT133				DYTCMWDSSRPTKWF	26%	37%	-5 (FR1)/ +3 (FR3)		
PGT134				DYTCMWDSSRPTKWF	27%	39%	-5 (FR1)/ +3 (FR3)		

Figure 1. Clonal lineage maturation tree for the PGT121 family of broadly neutralizing antibodies, illustrating somatic hypermutation paths starting from the germline precursor (UCA) through ancestral intermediates to mature variants (reproduced from Sok *et al.* [29]).

portion of viral strains neutralized) directly from the antibody’s variable heavy-chain (V_H) sequence is challenging because the functional target depends on a highly diverse sequence, and laboratories measure it using inconsistent virus panels[add cite].

Protein language models (PLMs) utilize transformer architectures to learn contextual, high-dimensional sequence representations from millions of unlabeled proteins. The state-of-the-art multimodal PLM ESM3 learns through self-supervision to reason over protein sequence, 3D structure, and function [12, 13]. This raises two questions. First, can the general-protein representations that ESM3 learns support out-of-lineage bnAb modeling (predicting properties for unseen antibody families) using only the heavy-chain

sequence? Second, does unsupervised domain adaptation to antibody repertoires (using sequence-only data from the Observed Antibody Space [OAS] database [23]) improve this representation transfer?

We guard against data leakage because antibody datasets contain many somatically related variants; thus, a naive random split can place members of the same clonal lineage in both training and validation folds. This allows models to train on ancestors or descendants, rather than learning generalizable properties. In this project, we implement a metadata-derived lineage grouping for lineage isolation across folds. Because the study uses the same dataset to compare multiple model architectures (heads) and pooling methods, the cross-validation score represents model-selection estimates.

We organize the study around three contributions. First, we explain the metadata-derived lineage grouping and group-resampled uncertainty analysis for classification[is this novel??] and regression on the Compile/Analyze Tool for Neutralization Assay Data (CATNAP) database [30]. Second, we present a matched evaluation of sequence features, frozen ESM3, and OAS-tuned ESM3 (via LoRA) under identical downstream folds and heads. Third, we perform a controlled analysis of VAE interpolation on residue-resolution sequence embeddings. This separates observations from unsupported manifold claims.

2. Related Work

2.1. HIV bnAb measurement and development

CATNAP standardizes antibody-by-virus neutralization panels and associated metadata [30], curating IC50 and IC80 measurements from pseudovirus neutralization assays across antibody-virus pairs and providing the consistent numerical labels for our regression target. Longitudinal lineage studies show that breadth can change substantially as somatic variants accumulate, including within the CAP256/VRC26 lineage [1, 10], where members share high sequence identity yet span a wide range of neutralization phenotypes. These observations motivate lineage-held-out evaluation: high sequence similarity does not imply identical breadth, and lineage membership is itself a strong short-cut that a naive random split that a model could exploit without learning generalizable features.

2.2. Machine learning for HIV neutralization

Most computational work on HIV bnAbs predicts *virus* resistance to a fixed antibody panel rather than scoring antibody breadth from sequence; this review surveys this literature [6]. In this framing the model input is typically viral envelope[is viral envelope explained?] features and the output is a binary or continuous resistance label for a specific antibody. Not as useful for ranking which antibody

to deploy for diverse populations. Some recent methods sharpen the virus centered view. Multi-task learning across antibodies mitigates HIV-1 subtype bias and captures co-resistance patterns shared across the antibody panel [15], and BRAVE[was the BRAVE acronym written?] couples ESM2 embeddings of both antibody and virus sequences with a random-forest classifier to predict resistance across 33 bnAbs [11]. We evaluate the converse, antibody centered task, if ESM3 embeddings provide a richer representation antibody neutralization breadth, from their heavy-chain sequences.

2.3. Protein and antibody language models

ESM2 showed large-scale protein representation learning via masked language modeling on evolutionary-scale sequence corpora, yielding embeddings competitive with structure prediction benchmarks [18]. ESM3 extended this approach across protein sequence, structure, and function features. through generative, transformer, with a robust masking algorithm. The novel feature modeling enabled joint reasoning over residue, 3D coordinates, and functional annotations [13]. In contrast, antibody models use sequence corpora tailored to the unusual statistics of the immunoglobulin variable regions. AntiBERTy was trained it on hundreds of millions of natural antibody sequences, this and showed the representation space can cover maturation trajectories [27].[was this shhown?] AbLang trained separate models for heavy and light chains using OAS, reflecting the distinct mutational statistics of each chain, and demonstrates an antibody-specific advantage for sequence restoration [24]. More recent antibody-specific models scale further and target known biases: IgBert and IgT5 use over two billion unpaired and two million paired OAS sequences [16], while AbLang-2 targets germline bias, where antibody models tend to be dominated by proximity to the germline sequence rather than functional features that might not generalize to downstream tasks [25]. Observed Antibody Space provides clean and annotated repertoire sequences used in pre-training and often tuning [23].

2.4. General versus domain-specific representations

Whether antibody-specific pretraining is needed is unclear. Across diverse immune tasks, general protein language models can match or exceed domain-specific models when comparisons control for model size, training corpus scale, and layer selection [8]. This result is non-obvious because antibody sequences violate the distribution of general protein data, they are defined by the somatic hypermutation and V(D)J recombinations that produce sequences rare or absent from other sets. Large general models may compensate through rich, learned representation. We prop this application of general protein models such as ESM3 against antibody feature extracted from the sequences. We compare

adaptation on the functional task of neutralization breadth, rather than a structural task tested in prior surveys.

2.5. Adaptation and latent interpolation

Low-Rank Adaptation (LoRA) allows tuning of a frozen model through trainable low-rank decompositions of the attention weights. This freezes over 99% of parameters allowing some adaptation a new/shifted distribution [14]. We use it on the masked self-supervised decoding task to tune ESM3 on OAS antibody sequences. We do not use LoRA for tuning on bnAb samples, thus, theoretically ESM3 is tuned for antibody sequence. Variational autoencoders (VAEs) [17] learn a regularized Gaussian latent space via a KL-divergence penalty on the autoencoder. This learned latent representation with regularization theoretically allows linear paths in the embedded space to correspond to semantically smooth interpolations between data points, creating the manifold. However, whether this holds for a given modality depends on the representation quality, decoder fidelity, and of course, the quality and amount of data. Our VAE experiment compares latent interpolation with a straight-line in the VAE latent space, with a similar interpolation in the ESM3 embedding space.

3. Data and Evaluation Protocol

3.1. CATNAP cohort and labels

We compile a dataset from the CATNAP database, which aggregates antibody sequences and quantitative viral neutralizations. The half-maximal inhibitory concentration (IC_{50}) measures an antibody’s potency against a specific viral isolate: it gives the concentration of antibody needed to neutralize 50% of the virus (lower values indicate higher antibody potency). We evaluate each antibody–virus as neutralized, if it is neutralized at the resistance threshold of $50 \mu\text{g/mL}$. Neutralization breadth b_i , for an antibody i , is the ratio of of tested antibody–virus measurements neutralized below the threshold, also given here:

$$b_i = \frac{\#\{\text{measurements with } IC_{50} \leq 50 \mu\text{g/mL}\}}{\#\{\text{tested antibody–virus measurements}\}}, \quad (1)$$

We restrict our analysis to antibodies tested against at least 20 virus strains for decreased variance in selection. For a binary classification task, we assign the “broad” label (1) where $b_i \geq 0.50$, and the narrow/weak label (0) where $b_i \leq 0.20$. We exclude the middle band to establish a distinct boundary between the binary classes. [cite a reason for the choice of the middle band region, size] We check structural variable-heavy-region (V_H) completeness by requiring conserved cysteine residues and framework markers to isolate the variable domain[explain this, and CDR in the intro.] (Figure 2 shows the V_H domain within the overall antibody structure) while retaining somatic mutations.

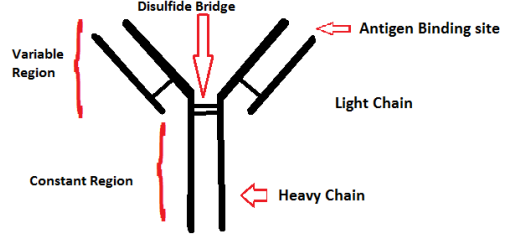


Figure 2. Schematic diagram of IgG antibody structure, highlighting the heavy and light chains, variable and constant domains, antigen-binding site, and disulfide bonds (reproduced from Mamahdi14 [21]).

The final labeled set contains the 300 unique V_H sequences (202 broad and 98 weak), with sequence lengths ranging from 116 to 146 amino acids. For continuous-breadth regression, we predict the ratio b_i directly for the 300 selected assays.

3.2. Lineage grouping and split

To prevent data leakage of closely related sequences, we group antibodies by distinct lineages, we do a robust hierarchical sorting to ensure lineage stratification. The hierarchical sort uses CATNAP clonal-lineage metadata (216 antibodies), patient metadata indicating the same donor (70), sequence-level clustering via single-linkage clustering at a threshold of $\geq 80\%$ Levenshtein sequence identity (2 antibodies), and the remaining individual singletons (12). This produces 113 unique group IDs. The tiers are not cross-merged, leaving nine near-duplicate pairs that span a group boundary, which we audit but intentionally do not merge (Appendix A.4). Across 774 in-group pairs, median pairwise sequence identity is 0.779 (Interquartile Range [IQR] 0.704–0.885); across 44,076 out-group pairs, the median is 0.424 (IQR 0.377–0.504).

Both classification and regression tasks utilize a single 5-fold StratifiedGroupKFold split. This partitioning scheme guarantees that all antibodies belonging to any of the 113 lineage groups remain together in either the training or the validation set (unseen group evaluation), while simultaneously stratifying folds to maintain the overall binary label ratio (broad vs. narrow) in each split.

3.3. Metrics, selection, and uncertainty

We evaluate classification performance using the Area Under the Receiver Operating Characteristic curve (AUROC) and Area Under the Precision-Recall Curve (AUPRC, or average precision). AUROC measures the probability that the classifier ranks a random positive instance higher than a random negative one. AUPRC evaluates the precision-recall trade-off, this is informative under class imbalances. For regression on the continuous breadth metric, we report: the

Pearson correlation coefficient (r), which measures linear correlation; the Spearman rank correlation (ρ), which evaluates monotonic but non-linear relationships; and the coefficient of determination (R^2), representing the proportion of target variance explained by the model compared to a simple mean baseline.

For hyperparameter tuning and configuration selection, we use the mean score across the 5 validation folds (mean AUROC for classification; mean Spearman ρ for regression). After selecting a configuration, we report the primary performance metrics on pooled out-of-fold (OOF) predictions (concatenating the held-out predictions from all folds). This pooling prevents bias from differing fold sizes and class compositions. To estimate prediction variance, we calculate 95% uncertainty intervals using 10,000 bootstrap draws, resampling complete lineage groups with replacement rather than individual sequences. To compute paired deltas that evaluate model differences, we resample the groups and calculate the metric difference on aligned OOF predictions. These intervals quantify sensitivity to lineage composition but do not account for configuration-selection bias.

3.4. OAS adaptation corpus

The executed domain-adaptation run uses 219,837 globally deduplicated human heavy-chain sequences from 11 OAS data units (108,675 clonotypes), with the largest unit contributing 20,000 sequences (9.1%). Whole data units that are systematically 5' truncated do not enter the corpus. A clonotype-aware split produces 198,130 training and 21,707 validation sequences. OAS supplies repertoire data for adaptation only; we neither label OAS sequences as HIV bnAbs nor include them in downstream test folds [23].

3.5. Lineage-interpolation cohort

The interpolation track unions the 300 labeled antibodies with 11 trajectory rows; four trajectory IDs already occur in the labeled cohort, yielding 307 unique sequences and 39,129 residue embeddings. The VAE training split contains 214 antibodies (27,490 residues), a group-held validation split contains 46 antibodies (5,738 residues), and the complete VRC34, VRC01, DH270, and DH511 lineages form the holdout set (47 antibodies; 5,901 residues).

4. Representations and Models

4.1. Handcrafted baseline

The baseline represents each V_H sequence with seven features: length; charged, hydrophobic, aromatic, and cysteine fractions; cysteine count; and N -glycosylation-motif count. We evaluate logistic regression, random forest [2], and XGBoost [3] heads. We select XGBoost as the baseline and

retain a length-only logistic regression as a simple diagnostic.

4.2. Frozen ESM3 and downstream heads

We represent each sequence with residue embeddings from the frozen, open-weights `esm3-sm-open-v1` model [12]. ESM3 is a 1.4-billion-parameter transformer model that learns from public protein databases. We pass each amino acid sequence through the model, extract the 1536-dimensional representation for each residue (excluding start-of-sequence BOS and end-of-sequence EOS tokens), resulting in a matrix of size $L \times 1536$ for a sequence of length L . To convert these variable-length representations into a fixed-size feature vector for downstream modeling, we compare two pooling schemes: (1) simple mean pooling across the sequence length, yielding a 1536-dimensional vector, and (2) a concatenated representation of the mean, maximum, and standard deviation across the sequence, yielding a 4608-dimensional vector. Downstream models fit their preprocessing steps (standard scaling and Principal Component Analysis [PCA] for dimensionality reduction) strictly inside each training fold to avoid target leakage. All downstream classifier and regressor heads, scaling, and PCA steps use the scikit-learn library [26].

Classification candidates are L1/L2 logistic regression (using the SAGA solver [7] for L1), random forest [2], XGBoost, and a PCA-50 multilayer perceptron. A separately trained additive attention-pooling classifier is an additional capacity test. Regression candidates are Ridge, random forest [2], and XGBoost, with a length-only Ridge floor. Appendix A.1 lists the hyperparameters.

4.3. OAS LoRA adaptation

To align general-protein representations with the specific statistical distribution of human antibodies, we perform domain adaptation using Low-Rank Adaptation (LoRA) [14]. LoRA freezes the base model parameters and injects trainable low-rank decomposition matrices into the transformer’s attention and feed-forward layers, drastically reducing the parameter space. We configure rank-16 adapters ($\alpha = 32$, dropout 0.05), training 18,874,368 parameters out of 1.42 billion total parameters (1.33%). Adaptation uses a self-supervised Masked Language Modeling (MLM) objective [9]: we randomly mask or corrupt 15% of amino acid tokens, and the model learns to predict the original residues from the surrounding context. We train for one epoch using the AdamW optimizer [20] with an effective batch size of 32. Performance evaluation uses cross-entropy loss and perplexity (the exponential of the loss, representing the effective branching factor when predicting residues). Our run achieves a clonotype-held-out validation loss of 0.5025 and a perplexity of 1.653. We merge the learned adapter weights directly into the base model parameters before extracting

embeddings.

4.4. Per-residue VAE and controls

To analyze sequence maturation and interpolation, we train a Variational Autoencoder (VAE) to compress 1536-dimensional residue-level embeddings into a smooth, lower-dimensional latent space. Variational autoencoders are generative models consisting of an encoder that maps input vectors to a latent probability distribution, and a decoder that reconstructs the inputs from samples drawn from this distribution. Our VAE architecture uses fully connected layers with hidden dimensions of 512 and 256, compressing the input to a 32-dimensional latent variable z . The loss function balances reconstruction error (Mean Squared Error) with a regularization term (Kullback-Leibler [KL] divergence) that penalizes deviation from a standard normal prior distribution to ensure a smooth latent space. We scale the KL penalty by $\beta = 0.005$ using a 25-epoch linear warm-up to prevent posterior collapse, wherein the model ignores the latent codes. We fit the standard scaler on training residue vectors only. We train separate frozen-embedding and LoRA-embedding VAEs with this same architecture for 300 epochs.

We globally align endpoint sequences with the Needleman-Wunsch algorithm [22] through the Biopython library [4] before residue-wise interpolation. For a UCA latent z_u and mature latent z_m , the VAE path is

$$z(\alpha) = (1 - \alpha)z_u + \alpha z_m, \quad \alpha \in [0, 1]. \quad (2)$$

We evaluate three embedding-space paths: the decoded VAE latent path; a straight line between VAE-reconstructed endpoints (`sl_rec`), which controls for reconstruction; and a straight line between raw ESM3 endpoints (`sl_raw`), which avoids the VAE bottleneck.

We perform four evaluations of the latent space geometry. The first evaluation (inferred intermediate search) searches an 11-step VRC34 UCA-to-mature path for the closest point to each of seven inferred intermediates (inferred ancestral sequences representing discrete evolutionary steps during maturation). The second evaluation (maturation midpoint test) tests four exact midpoints of ordered VRC34 triples. The third evaluation (maturation correlation analysis) correlates a VAE latent-axis coordinate with the sequence edit distance (the number of single-amino-acid substitutions, insertions, or deletions) from the UCA for four held-out lineages, serving as a proxy for evolutionary time. The fourth evaluation (latent space geometry check) compares VAE and reconstructed-line midpoints (geometric centers of the interpolation paths) for 75 sampled triples across 15 lineages to evaluate latent space smoothness. The maturation correlation analysis lacks a raw-embedding projection control and therefore cannot isolate a VAE or LoRA effect.

Representation	Fold AUROC	Pooled AUROC	95% interval
Handcrafted XGB	0.644 $\pm$ 0.086	0.625	[0.518, 0.744]
Frozen ESM3	0.761 $\pm$ 0.131	0.746	[0.653, 0.837]
OAS LoRA ESM3	0.745 $\pm$ 0.140	0.726	[0.626, 0.825]

Table 1. Classification performance for the selected handcrafted, frozen, and LoRA ESM3 configurations. A group bootstrap produces the intervals.

5. Results

5.1. Classification

Table 1 separates the configuration-selection score (mean over folds) from the pooled OOF estimate and group-bootstrap interval. Frozen ESM3 improves pooled AUROC from 0.625 to 0.746 and pooled AUPRC from 0.742 to 0.859. On aligned predictions, the paired frozen-minus-handcrafted difference is +0.121 AUROC ([0.012, 0.228]) and +0.117 AUPRC ([0.028, 0.207]). The intervals describe sensitivity to held-out lineage composition; they do not remove model-selection optimism.

LoRA’s selected classifier reaches pooled AUROC 0.726, below frozen by -0.020 with paired interval $[-0.096, 0.059]$. The corresponding AUPRC difference is -0.031 $[-0.075, 0.024]$. These data do not establish a downstream classification improvement from the single adaptation run. The receiver operating characteristic and precision-recall curves for these classifiers appear in Figure 3.

The learned attention pool reaches mean-fold AUROC 0.702 ± 0.100 , below the best fixed-pooling frozen model. This is a negative capacity result on the observed cohort, not evidence that learned pooling is generally inferior.

5.2. Continuous breadth regression

Regression uses the same 300 threshold-selected antibodies. As summarized in Table 2, frozen ESM3 reaches pooled Spearman $\rho = 0.366$ ([0.126, 0.585]), Pearson $r = 0.357$ ([0.129, 0.580]), and $R^2 = 0.114$ ($[-0.134, 0.313]$). The length-only pooled Spearman floor is 0.004. Thus the se-

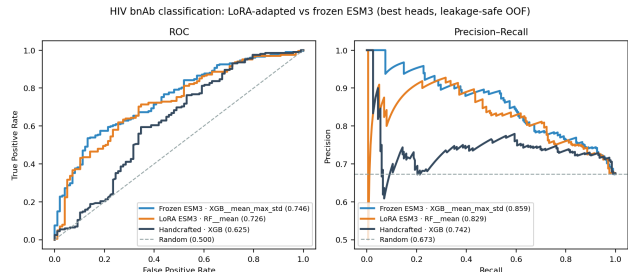


Figure 3. Pooled OOF ROC and precision-recall curves for the selected classifiers.

Representation	Fold ρ	Pooled ρ	95% interval
Frozen ESM3	0.380 ± 0.312	0.366	[0.126, 0.585]
OAS LoRA ESM3	0.357 ± 0.320	0.319	[0.059, 0.561]
Length only	—	0.004	—

Table 2. Continuous-breadth regression performance. Pooled OOF metrics and group-bootstrap intervals evaluate the selected configurations.

lected representation contains ranking information, while its absolute breadth predictions remain weak and sensitive to lineage composition. The scatter plots of predicted versus true breadth values appear in Figure 4, illustrating the prediction variance.

Selecting LoRA configurations by mean-fold Spearman chooses random forest with mean pooling, not the previously pooled-selected XGBoost configuration. Its pooled Spearman is 0.319 ([0.059, 0.561]). The paired LoRA-minus-frozen Spearman difference is -0.047 ($[-0.169, 0.064]$).

5.3. Reconstruction dominates VAE interpolation

The frozen-embedding VAE’s best epoch has validation total loss 0.2599, reconstruction loss 0.1859, and KL loss 14.80. These values document the saved run; they do not by themselves establish useful interpolation geometry.

In the inferred intermediate search (detailed in Table 3), raw ESM3 interpolation is closer than the VAE path for six of seven VRC34 inferred intermediates. Raw-line Mean Absolute Error (MAE) ranges from 8.51 to 46.55 (mean 20.77), while VAE-path MAE ranges from 34.54 to 44.30 (mean 37.47). The exception is $pI3'$, for which the raw-line MAE is 46.55 and the VAE MAE is 44.30. The VAE-minus-sl_rec difference averages only -0.0069 (range -0.0196 to approximately zero), so the saved case study does not demonstrate practically meaningful curvature beyond reconstruction.

The maturation midpoint test provides the cleanest direct



Figure 4. Pooled OOF predicted versus observed breadth for frozen and LoRA configurations.

Inferred intermediate	VAE MAE	Raw-line MAE	VAE-rec. line
$pI1$	37.96	8.51	-0.0000
$pI2$	34.54	15.54	-0.0106
$pI2'$	38.28	32.48	-0.0196
$pI2$ H33P	35.29	17.14	-0.0179
$pI3$	36.56	15.53	0.0000
$pI3'$	44.30	46.55	-0.0000
$pI4$	35.38	9.63	-0.0000

Table 3. Exact embedding-space MAE per dimension/residue for VRC34 intermediates (inferred intermediate search). The reconstructed straight-line control provides the comparison for VAE performance.

midpoint comparison. Across four ordered VRC34 triples, raw-line midpoint MAE ranges from 8.24 to 13.35, versus 35.20 to 37.95 for the VAE midpoint. The per-residue VAE bottleneck therefore adds much more error than the midpoint-control differences under study.

5.4. LoRA and descriptive maturation directionality

For the maturation correlation analysis, mean Pearson correlation across the four held-out lineages increases from 0.566 for the frozen-embedding VAE to 0.822 for the LoRA-embedding VAE; mean Spearman increases from 0.384 to 0.692. All four Pearson point estimates increase (Figure 5), but sensitivity intervals are wide for small lineages. For example, DH511 changes from 0.084 to 0.409, while its approximate LoRA Pearson interval is $[-0.415, 0.865]$.

This comparison is descriptive. It contrasts two separately trained, single-seed VAEs, uses one LoRA adapter, covers four lineages, and lacks a raw-embedding projection control. It therefore cannot establish that OAS adaptation caused improved maturation organization.

The latent space geometry check does not provide supporting interpolation evidence. Frozen VAE-minus-line MAE has mean 0.041, standard deviation 0.274, and lineage-bootstrap interval $[-0.016, 0.109]$; LoRA has mean 0.076, standard deviation 0.237, and interval $[0.023, 0.142]$. Positive differences favor the reconstructed straight-line control.

6. Limitations

6.1. Exploratory model selection

We compare heads and pooling schemes on the same five grouped folds that we use for reporting. Selecting the best mean-fold configuration makes its reported cross-validation performance optimistic due to hyperparameter and model-selection leak. Group-bootstrap intervals condition on the selected predictions and do not account for that selection process. A future confirmatory evaluation requires nested

grouped cross-validation or a fully locked external lineage holdout.

6.2. Cohort and target construction

The dataset is extremely small (300 sequences total across 113 lineage groups), making downstream models highly susceptible to overfitting and limiting the statistical power of our architectural comparisons. Only variable heavy chains (V_H) enter the models, so the analysis completely neglects light-chain (V_L) contributions and heavy-light pairing, which are biochemically known to be critical for the folding, stability, and neutralization breadth of many HIV-1 bnAbs. Furthermore, CATNAP narrow/weak antibodies are not equivalent to arbitrary non-bnAbs, and the analysis treats neutralization breadth as a simple, unweighted fraction over viral panels that vary in size and strain composition, without accounting for laboratory-specific or panel-level effects.

6.3. Grouping assumptions

While metadata-derived clonal lineage is the strongest available grouping key, patient-level fallback groups may merge unrelated antibodies from the same donor, while singletons may mask unrecorded genetic relationships. The sequence-identity clustering heuristic ($\geq 80\%$ identity) supports separation in aggregate but cannot biologically prove complete developmental independence or rule out convergent evolution.

6.4. Adaptation and ESM3 constraints

We evaluate the LoRA domain adaptation experiment on a single dataset with one adapter configuration and one training run. Because the adaptation is unsupervised (using unlabeled OAS sequences), the model may align representations toward general repertoire characteristics at the expense of capturing specific somatic hypermutation features required for HIV-1 gp120/gp41 binding. Additionally, using a 1.4B

parameter model like ESM3 introduces substantial computational overhead for inference and adaptation compared to simpler sequence-based models, and its non-commercial licensing constraints limit downstream translation.

6.5. Interpolation and VAE constraints

The per-residue VAE treats residues as independent training examples and lacks an antibody-level encoder. By treating residues independently, the model ignores sequence order, tertiary structural contacts, and global cooperative folding constraints. This constitutes a severe biophysical simplification. Furthermore, the VAE failed our round-trip sequence decoding test-gate, meaning the latent spaces learned at the residue level cannot be reliably mapped back to constructible antibody sequences. Any claims of sequence design or maturation interpolation are therefore speculative and physically unvalidated.

6.6. Reproducibility boundary

Our verification script validates pre-computed predictions and fails on cohort or alignment mismatches, but the massive embedding caches and LoRA adapter weights come from external GPU clusters (Google Colab), and the local verification run does not recreate them. The saved prediction and evaluation artifacts therefore reproduce the paper’s claims, but a full, local, from-scratch retraining pipeline does not.

7. Conclusion

With lineage-grouped evaluation, frozen ESM3 representations show an improvement over sequence features for CATNAP bnAb classification and show a modest increase in breadth performance. However, absolute breadth prediction remains weak, indicating that heavy-chain sequence features alone, even from large PLMs, are insufficient for accurately modeling functional neutralization breadth. Furthermore, our unsupervised OAS LoRA adaptation run did not improve downstream performance, suggesting that general-repertoire adaptation without target-specific supervision is insufficient for capturing the complex somatic hypermutation patterns critical to HIV-1 neutralization.

To our knowledge the framing itself is novel: prior machine learning for HIV neutralization is predominantly *virus-centric*, scoring envelope resistance against a fixed antibody panel [6, 11, 15], and the closest antibody-centric breadth estimator infers breadth by molecular modeling onto a template antibody-antigen complex for a single epitope class [5]. We instead classify breadth directly from heavy-chain sequence (rather than from a modeled structure), epitope-agnostically, and under a lineage-held-out split. In that out-of-lineage setting frozen ESM3 raises pooled AUROC from 0.625 for a handcrafted biophysical

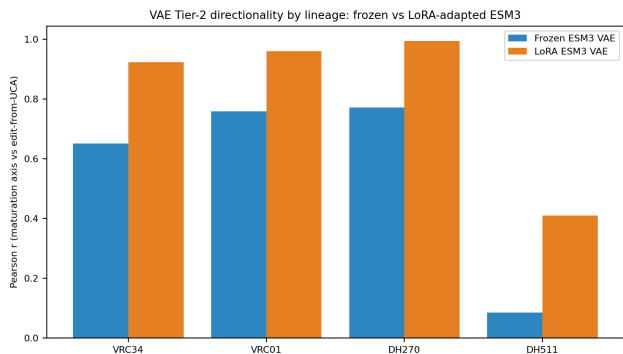


Figure 5. Pearson maturation correlation for four held-out lineages (maturation correlation analysis).

baseline to 0.746 (paired +0.121, group-bootstrap interval [0.012, 0.228] excluding zero). Because no published benchmark shares this sequence-only, epitope-agnostic, lineage-held-out protocol, external AUROCs from random-split or virus-side tasks are not directly comparable; the meaningful baseline is our own handcrafted-feature model, over which ESM3 shows a credible out-of-lineage gain.

The residue-level VAE interpolation analysis shows cautionary results. The per-residue VAE model introduces substantial reconstruction error and fails to outperform simpler midpoint controls on the VRC34 lineage, likely due to the obvious lack of bnAb data. Due to that fact, we cannot draw strong conclusions on whether a VAE would meaningfully support the generation or design of functional antibodies, or about VAE-space interpolation. While the LoRA maturation correlations were higher, the limited exploration of this tuning does not establish a negative causal effect.

Ultimately, our strongest contribution is methodological; implementing lineage-aware splits, pooled-versus-fold validation reporting, and group-resampled uncertainty estimation changes the interpretation of these experiments.

Our results motivate natural next steps. Because heavy-chain sequence alone proved insufficient for breadth, richer inputs such as paired heavy/light chains or explicit structure are worth exploring. Additionally, the results of the VAE and adaptation analyses were limited primarily by the scarcity of labeled bnAb data, larger cohorts and external validation on held-out lineages are needed before stronger conclusions can be drawn.

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A. Verification and Reproducibility Details

A.1. Candidate configurations

Table 4 records the compared model families and the fixed settings that materially affect interpretation, including Logistic Regression (LR), Random Forest (RF), XGBoost (XGB), Multilayer Perceptron (MLP), and Principal Component Analysis (PCA). Figure 6 shows the relative importances of the handcrafted biophysical features in the baseline XGBoost classifier.

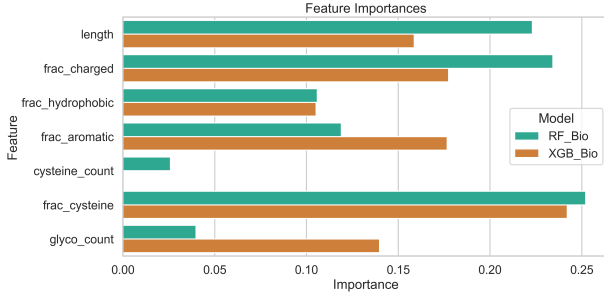


Figure 6. Handcrafted feature importances for the baseline XGBoost classifier.

Component	Candidates and fixed settings
Handcrafted classifier	LR, RF, XGB; seven features; RF/XGB depth 3; grouped five-fold CV
Frozen/LoRA classifier	L1/L2 LR, RF, XGB, MLP; mean or mean+max+std pooling; RF 300 trees/depth 3; XGB 200 trees/depth 3; MLP after PCA-50
Attention classifier	Additive token attention; masked pooling, dropout 0.4, weight decay 10^{-2} , 120 epochs
Frozen/LoRA regressor	Ridge, RF, XGB; mean or mean+max+std; RF 300 trees/depth 4; XGB 200 trees/depth 3
OAS adaptation	One rank-16 LoRA run; $\alpha = 32$, dropout 0.05, learning rate 10^{-4} , one epoch, effective batch 32
Per-residue VAE	One run per representation; 1536–512–256–32 encoder, mirrored decoder, dropout 0.1, $\beta = 0.005$, 25-epoch warm-up, 300 epochs

Table 4. Evaluated configurations. Because these candidates share reporting folds, selected cross-validation estimates remain exploratory.

A.2. Classification detail

Table 5 reconciles the configuration selection score with the pooled estimates used in the main text.

Representation	AUROC	AUPRC	Fold AUROC
Handcrafted XGB	0.625	0.742	0.644 ± 0.086
Frozen ESM3 XGB	0.746	0.859	0.761 ± 0.131
OAS LoRA ESM3 RF	0.726	0.829	0.745 ± 0.140

Table 5. Classification metrics reconciled from aligned OOF predictions. Frozen XGB uses mean+max+std pooling; LoRA RF uses mean pooling. Configurations maximize mean-fold AUROC.

A.3. Regression detail

The fixed, mean-fold-selected regression configurations give the pooled estimates in Table 6.

Representation	Spearman	Pearson	R^2
Frozen ESM3	0.366	0.357	0.114
OAS LoRA ESM3	0.319	0.310	0.036
Length-only Ridge	0.004	−0.059	−0.046

Table 6. Pooled OOF breadth-regression metrics. Frozen and LoRA configurations maximize mean-fold Spearman.

A.4. Lineage-group leakage audit

The fallback grouping (Section 3) does not cross-merge tiers, so an antibody can resemble a member of a different group. We surface every such case with `find_cross_group_near_duplicates`: across all $\binom{300}{2}$ antibody pairs, nine span a group boundary while exceeding the 80% Levenshtein-identity threshold (Table 7). Several involve inferred ancestral sequences (unmutated common ancestors and intermediates, e.g. VRC34–UCA, DH270 UCA/IA4) or germline-reverted variants that share a V_H germline across lineages or donors. They are deliberately *not* auto-merged; the count quantifies the residual leakage left after grouping rather than eliminating it silently.

Antibody A	Antibody B	Group A	Group B	Id.
1361	D5_AR	patient:donor.Gormy2012	cluster:10	0.811
1361	PGZL1.gVmDmJ	patient:donor.Gormy2012	lineage:PGZL1	0.839
1393A	D5_AR	patient:donor.Gormy2012	cluster:10	0.803
1393A	PGZL1.gVmDmJ	patient:donor.Gormy2012	lineage:PGZL1	0.855
D5_AR	PGZL1.gVmDmJ	cluster:10	lineage:PGZL1	0.815
D5_FI	PGZL1.gVmDmJ	cluster:10	lineage:PGZL1	0.806
DH270 IA4	VRC34-UCA	lineage:DH270	lineage:VRC34	0.843
DH270 UCA	VRC34-UCA	lineage:DH270	lineage:VRC34	0.874
PGDM1400	PGDM1401	lineage:PGDM1400	patient:Donor 84	0.816

Table 7. Residual cross-group near-duplicates: antibody pairs with $\geq 80\%$ Levenshtein identity assigned to different lineage groups. Group labels carry their fallback tier (lineage, patient, cluster). These pairs are intentionally not merged, so the table measures the leakage remaining after the hierarchical grouping.

A.5. Maturation correlation sensitivity

Small lineage sizes make the Fisher- z sensitivity intervals in Table 8 wide, especially for DH270 and DH511.

A.6. Interpretation rules

The paper uses “interval” rather than “confidence interval” where the resampling assumptions are especially limited. It avoids significance claims for the VAE evaluations, does not treat non-overlap or interval sign as proof of a biological mechanism, and distinguishes direct observations from causal interpretation. Project history retains random-split

Lineage	n	Frozen r [interval]	LoRA r [interval]
VRC34	14	0.651 [0.184, 0.878]	0.923 [0.768, 0.976]
VRC01	15	0.759 [0.404, 0.915]	0.960 [0.881, 0.987]
DH270	6	0.771 [-0.108, 0.973]	0.995 [0.950, 0.999]
DH511	8	0.084 [-0.660, 0.745]	0.409 [-0.415, 0.865]

Table 8. Descriptive Pearson correlations and Fisher- z sensitivity intervals (maturation correlation analysis). These are not independent estimates of a causal adaptation effect.

and earlier name-grouped scores, but the main result narrative excludes them because their current prediction artifacts are not part of the verified canonical bundle.

A.7. Per-configuration discriminative results

Tables 9 and 10 report every fixed-pooling configuration compared for the frozen and LoRA representations. The entries are not independent experiments: all candidates reuse the same five reporting folds. Selection uses the mean-fold column, even when a different configuration has a larger pooled estimate. The complete classification performance across all heads and pooling schemes appears in Figure 7, while Figure 8 details the regression Spearman correlations.

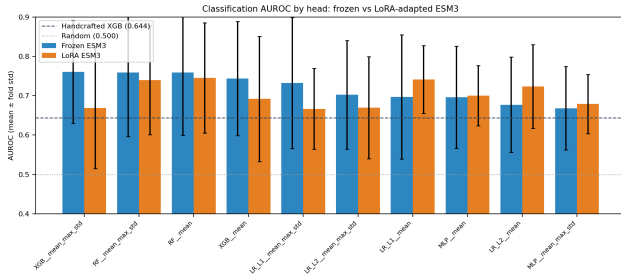


Figure 7. Mean grouped-fold AUROC (with standard deviation) for all evaluated classifier heads and pooling configurations under frozen and LoRA representations.

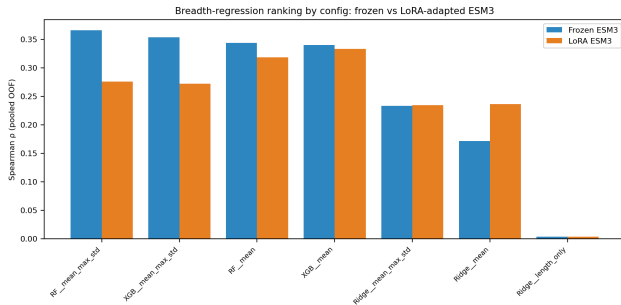


Figure 8. Regression Spearman correlation across different model configurations, comparing frozen and LoRA representations.

Configuration	Frozen		LoRA	
	AUROC	AUPRC	AUROC	AUPRC
XGB, mean+max+std	0.761	0.878	0.668	0.794
RF, mean+max+std	0.759	0.874	0.740	0.855
RF, mean	0.759	0.875	0.745	0.858
XGB, mean	0.744	0.864	0.692	0.808
L1 LR, mean+max+std	0.732	0.858	0.667	0.805
L2 LR, mean+max+std	0.702	0.836	0.670	0.797
L1 LR, mean	0.697	0.827	0.741	0.834
MLP, mean	0.696	0.836	0.700	0.803
L2 LR, mean	0.677	0.822	0.723	0.833
MLP, mean+max+std	0.668	0.804	0.679	0.783

Table 9. Mean grouped-fold classification results for all frozen and LoRA fixed-pooling candidates. Bold marks the selected LoRA configuration; the selected frozen configuration is the first row.

Configuration	Frozen Spearman		LoRA Spearman	
	Fold	Pooled	Fold	Pooled
RF, mean+max+std	0.380	0.366	0.304	0.276
RF, mean	0.342	0.344	0.357	0.319
XGB, mean+max+std	0.341	0.354	0.292	0.272
XGB, mean	0.329	0.340	0.353	0.333
Ridge, mean+max+std	0.200	0.233	0.226	0.235
Ridge, mean	0.149	0.171	0.208	0.236
Length-only Ridge	0.155	0.004	0.155	0.004

Table 10. Mean-fold and pooled-OOF regression Spearman correlations for all candidates. Bold marks the configurations selected by mean-fold Spearman.

A.8. Paired resampling results

The verification script aligns predictions by antibody ID and resamples complete `group_id` units. Table 11 gives the resulting paired differences; 10,000 valid resamples produce every interval.

Comparison	Metric	Δ	Low	High
Frozen-handcrafted	AUROC	0.121	0.012	0.228
Frozen-handcrafted	AUPRC	0.117	0.028	0.207
LoRA-frozen	AUROC	-0.020	-0.096	0.059
LoRA-frozen	AUPRC	-0.031	-0.075	0.024
LoRA-frozen	Spearman	-0.047	-0.169	0.064
LoRA-frozen	Pearson	-0.047	-0.173	0.071
LoRA-frozen	R^2	-0.078	-0.228	0.053

Table 11. Paired group-bootstrap differences for the selected configurations. Intervals quantify sensitivity to the observed lineage groups and do not correct for configuration selection.

A.9. VAE reconstruction and geometry summary

The two VAEs use identical split membership and architecture, but we train them in separate single runs. The frozen and LoRA best epochs are 299 and 296, respectively; their best validation total/reconstruction/KL losses are 0.2599/0.1859/14.80 and 0.2834/0.2045/15.79. Each run uses 27,490 training, 5,738 validation, and 5,901 held-out residue embeddings. The detailed VAE geometry checks and verification metrics appear in Table 12.

Verified VAE quantity	Frozen	LoRA
Mean Pearson correlation (maturation analysis)	0.566	0.822
Mean Spearman correlation (maturation analysis)	0.384	0.692
Mean VAE–line MAE (geometry check)	0.041	0.076
Standard deviation (geometry check)	0.274	0.237
Median (geometry check)	0.004	0.018
Positive fraction (geometry check)	0.573	0.693
Lineage bootstrap low (geometry check)	−0.016	0.023
Lineage bootstrap high (geometry check)	0.109	0.142

Table 12. Descriptive VAE geometry summaries. Positive VAE–line differences (geometry check) favor the reconstructed straight-line control.

A.10. Reproducibility limits

The CPU verification script reconstructs claims from saved OOF predictions and evaluation artifacts; it does not retrain ESM3, the LoRA adapter, or either VAE. Therefore, it verifies internal consistency and reported calculations, not bit-wise reproducibility of the GPU training runs. Similarly, the group bootstrap resamples the observed 113 metadata-derived groups. It does not represent a new external cohort, and its intervals remain conditional on the selected model configurations.

Confirmatory evidence would require a locked external lineage cohort or nested grouped model selection, repeated adapter and VAE seeds, and raw-embedding directionality controls for every lineage. Those missing experiments are limitations of the current evidence, not tasks silently replaced by the claim verification.